HW6

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Prob. 10

Since P is a nonzero projector, it satisfies the idempotency property by definition.

$$P^2 = P$$

Any eigenvalue λ of a projector must satisfy the equation $\lambda^2 = \lambda$, which implies that the only possible eigenvalues are $\lambda = 0$ or $\lambda = 1$.

Because P is given as a nonzero matrix, it must possess at least one eigenvalue equal to 1.

Recall that the spectral radius $\rho(P)$, defined as the maximum absolute value of the eigenvalues of P, provides a lower bound for the induced 2-norm of any matrix.

$$\|P\|_2 \geq \rho(P)$$

Substituting the known spectral radius into this inequality, we obtain the first part of the proof.

$$\|P\|_2 \geq 1$$

Now, we consider the condition for equality, $\|P\|_2 = 1$.

Suppose P is an orthogonal projector, which implies P is Hermitian (or symmetric in $\mathbb{R}^{m\times m}$), meaning $P = P^T$.

The singular values of P are defined as the square roots of the eigenvalues of $P^T P$.

For an orthogonal projector, we have the following identity.

$$P^T P = PP = P^2 = P$$

Since the eigenvalues of P are 1 and 0, the singular values of P are $\sqrt{1} = 1$ and $\sqrt{0} = 0$.

The 2-norm is the largest singular value, so $\|P\|_2 = 1$.

Conversely, suppose $\|P\|_2 = 1$.

Let the singular value decomposition (SVD) of P be given by the following expression.

$$P = U\Sigma V^T$$

In this decomposition, U and V are orthogonal matrices, and Σ is a diagonal matrix of singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$.

Given $\|P\|_2 = 1$, we have $\sigma_1 = 1$, and consequently, $1 \geq \sigma_i > 0$ for all $i \in \{1, \ldots, r\}$.

Substituting the SVD into the projector equation $P^2 = P$, we get the following.

$$(U\Sigma V^T)(U\Sigma V^T) = U\Sigma V^T$$

By multiplying by U^T on the left and V on the right, the equation simplifies.

$$\Sigma(V^T U)\Sigma = \Sigma$$

Let $W = V^T U$, where W is an orthogonal matrix.

The equation can be written in terms of its components for the first r indices.

$$\sigma_i w_{ii} \sigma_i = \sigma_i \implies w_{ii} = \frac{1}{\sigma_i}$$

Since $\sigma_i \leq 1$, it follows that $w_{ii} \geq 1$.

However, the entries of an orthogonal matrix must satisfy $|w_{ij}| \leq 1$, which forces $w_{ii} = 1$ and $\sigma_i = 1$ for all $i \in \{1, \ldots, r\}$.

Because the diagonal entries w_{ii} are all 1, the orthogonality of W requires that all other entries in the first r rows and columns must be zero.

$$W = \begin{pmatrix} I_r & 0 \\ 0 & W_{22} \end{pmatrix}$$

This structure implies $U = VW$.

Substituting this back into the SVD of P yields the following result.

$$P = VW\Sigma V^T = V \begin{pmatrix} I_r & 0 \\ 0 & W_{22} \end{pmatrix} \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} V^T = V \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} V^T$$

The resulting expression for P is clearly symmetric.

$$P^T = (V\Sigma V^T)^T = V\Sigma V^T = P$$

Since P is a symmetric projector, it is by definition an orthogonal projector.

Prob. 11

To solve the problem of computing the smallest singular value for the given $m \times m$ matrix A, we first define the structure of the matrix and the mathematical relationship between its singular values and eigenvalues.

The matrix A is defined as follows.

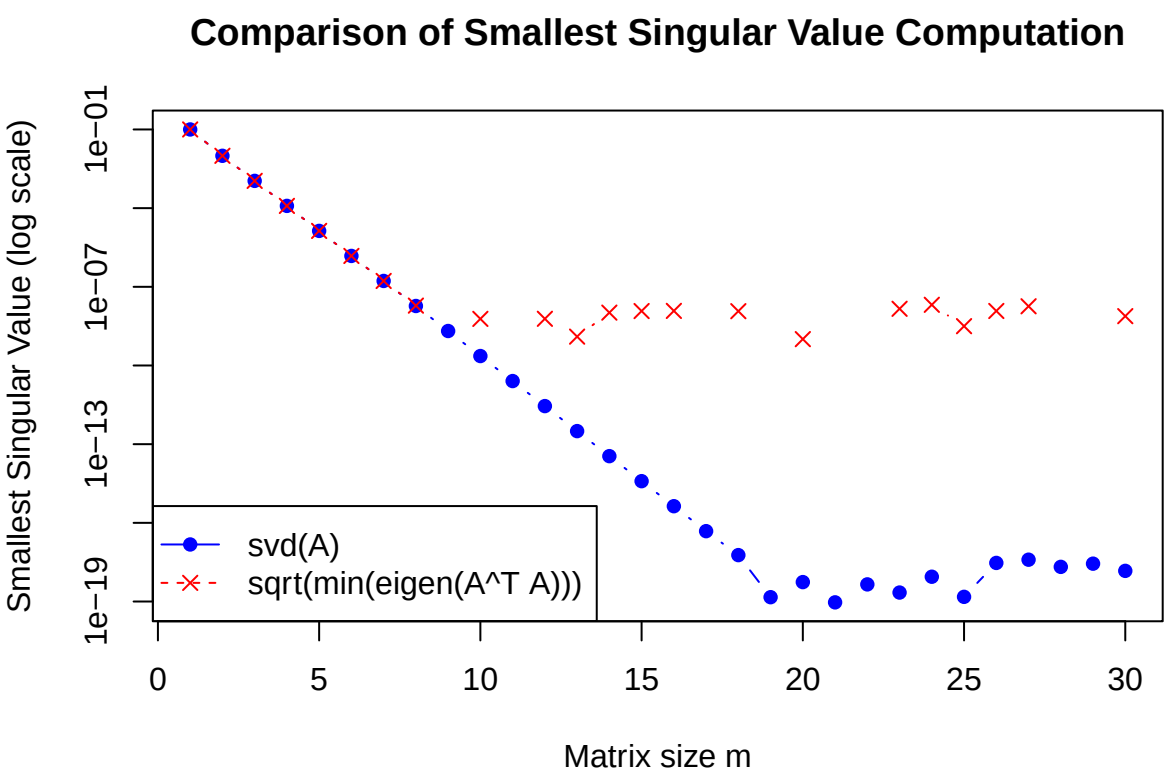
$$A = \begin{pmatrix} 0.1 & 1 & \ldots & 1 \\ 0 & 0.1 & \ldots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0.1 \end{pmatrix} \in \mathbb{R}^{m \times m}$$

The singular values σ_i of A are the square roots of the eigenvalues of the matrix $A^T A$.

$$\sigma_{min}(A) = \sqrt{\lambda_{min}(A^T A)}$$

The following R code implements both the direct SVD approach and the eigenvalue approach for $1 \leq m \leq 30$.

```r
m_values <- 1:30
svd_results <- numeric(30)
eigen_results <- numeric(30)

for (m in m_values) {
  A <- matrix(1, nrow = m, ncol = m)
  A[lower.tri(A)] <- 0
  diag(A) <- 0.1

  # Method 1: Direct SVD
  s_values <- svd(A)$d
  svd_results[m] <- min(s_values)
  # Method 2: Eigenvalues of A^T * A
  ata_eigen <- eigen(t(A) %*% A, symmetric = TRUE)$values
  # Ensure no negative values due to precision issues before sqrt
  eigen_results[m] <- sqrt(max(min(ata_eigen), 0))
}

# Plotting the results on a log scale
plot(m_values, svd_results, type = "b", col = "blue", log = "y", pch = 16,
     xlab = "Matrix size m", ylab = "Smallest Singular Value (log scale)",
     main = "Comparison of Smallest Singular Value Computation")
lines(m_values, eigen_results, type = "b", col = "red", pch = 4, lty = 2)
legend("bottomleft", legend = c("svd(A)", "sqrt(min(eigen(A^T A)))"),
       col = c("blue", "red"), pch = c(16, 4), lty = c(1, 2))
```



Prob. 12

The following R code constructs ten random matrices according to the specified distribution and computes the spectral abscissa alongside the matrix exponential norm over the interval $t \in [0, 20]$.

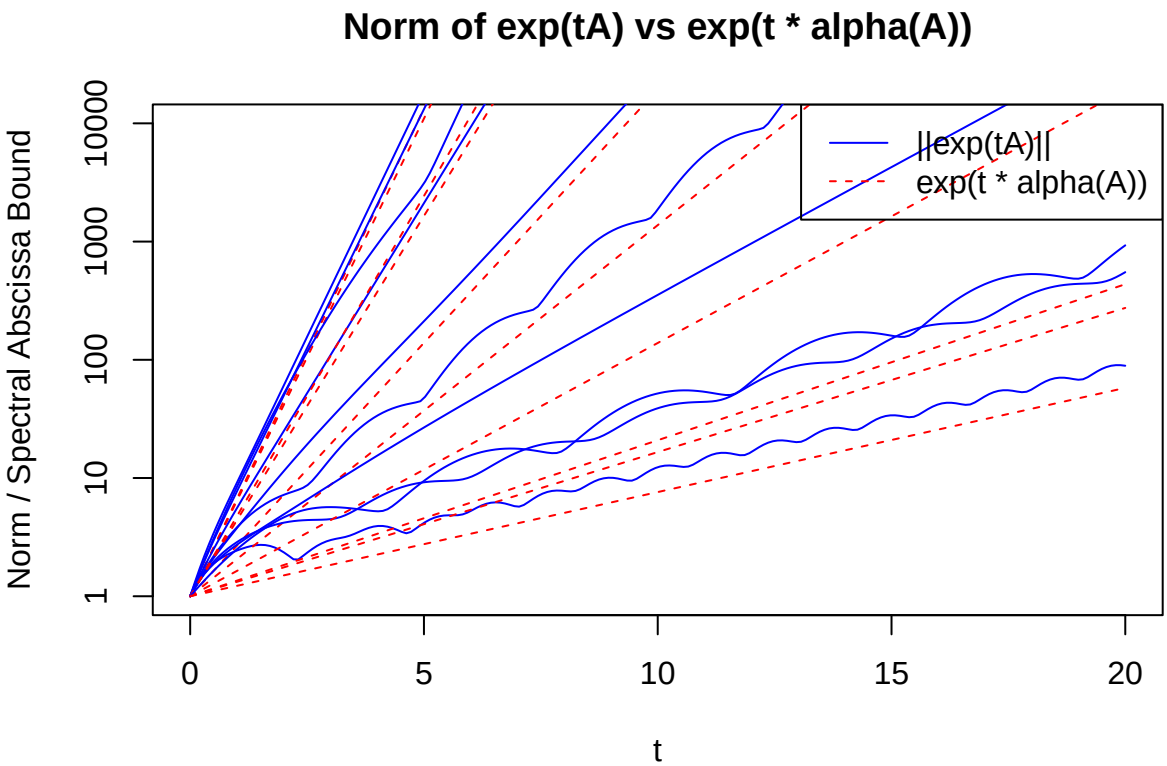
The matrix A is generated as $R - 2I$, where R consists of standard normal entries.

The library expm is utilized for the computation of the matrix exponential.

```r
library(expm)
```

```
## Loading required package: Matrix

##
## Attaching package: 'expm'

## The following object is masked from 'package:Matrix':
##
##     expm
```

```r
n <- 10
t_vals <- seq(0, 20, length.out = 200)
num_trials <- 10

plot(NULL, xlim = c(0, 20), ylim = c(1, 1e4), log = "y",
     xlab = "t", ylab = "Norm / Spectral Abscissa Bound",
     main = "Norm of exp(tA) vs exp(t * alpha(A))")

for (i in 1:num_trials) {
  A <- matrix(rnorm(n^2), nrow = n) - 2 * diag(n)

  # Calculate spectral abscissa alpha(A)
  eigs <- eigen(A, only.values = TRUE)$values
  alpha_A <- max(Re(eigs))

  # Compute ||exp(tA)|| and exp(t*alpha(A))
  norm_vals <- sapply(t_vals, function(t) norm(expm(t * A), type = "2"))
  bound_vals <- exp(t_vals * alpha_A)

  # Plot results
  lines(t_vals, norm_vals, col = "blue", lwd = 1)
  lines(t_vals, bound_vals, col = "red", lty = 2)
}

legend("topright", legend = c("||exp(tA)||", "exp(t * alpha(A))"),
       col = c("blue", "red"), lty = c(1, 2))
```



The spectral abscissa $\alpha(A)$ governs the long-term asymptotic behavior of the matrix exponential norm, though it often overlooks short-term dynamics.

$$\alpha(A) = \max_j \mathrm{Re}(\lambda_j)$$

In non-normal matrices, where $A^T A \neq AA^T$, non-orthogonal eigenvectors can cause transient growth, leading the norm to spike significantly before eventually decaying.

$$\|e^{tA}\|_2 \gg e^{t\alpha(A)} \text{ for small } t$$

If the spectral abscissa is determined by a complex conjugate pair of eigenvalues, the curve will exhibit persistent oscillations as $t \to \infty$.

$$\lambda = \alpha(A) \pm i\beta, \quad \beta \neq 0$$

The imaginary component β introduces sinusoidal fluctuations that prevent the norm from following a simple exponential path.

$$\|e^{tA}\|_2 \propto e^{t\alpha(A)} \left|\cos(\beta t + \phi)\right|$$

Conversely, if the dominant eigenvalue is purely real, the norm will eventually approach a smooth, monotonic exponential growth or decay.

$$\lim_{t\to\infty} \frac{\|e^{tA}\|_2}{e^{t\alpha(A)}} = C$$

Prob. 13

To find the eigenvalue decomposition of the block matrix B, we define the eigenvalue problem for a partitioned vector.

$$B \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$$

Expanding the matrix-vector product for the matrix $B = \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix}$, we obtain a system of coupled equations.

$$\begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} A^T y \\ Ax \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix}$$

This system implies two primary relations: $A^T y = \lambda x$ and $Ax = \lambda y$.

By substituting the second equation into the first, we can decouple the variables and identify the relationship to the singular values of A.

$$A^T(Ax) = A^T(\lambda y) = \lambda(A^T y) = \lambda^2 x$$

Similarly, substituting the first into the second yields the following.

$$A(A^T y) = A(\lambda x) = \lambda(Ax) = \lambda^2 y$$

These results demonstrate that x and y must be eigenvectors of $A^T A$ and AA^T, respectively, with the shared eigenvalue λ^2.

From the provided singular value decomposition $A = U\Sigma V^T$, we know that $A^T A = V\Sigma^2 V^T$ and $AA^T = U\Sigma^2 U^T$.

The eigenvalues of $A^T A$ are the squares of the singular values σ_i^2, meaning the eigenvalues of B satisfy $\lambda^2 = \sigma_i^2$, or $\lambda = \pm\sigma_i$.

The corresponding eigenvectors x and y are the right singular vectors v_i and left singular vectors u_i.

For each singular value $\sigma_i > 0$, we can construct two eigenvectors for B.

If $\lambda = \sigma_i$ and we set $x = v_i$, then $y = \frac{1}{\sigma_i} A v_i = u_i$.

If $\lambda = -\sigma_i$ and we set $x = v_i$, then $y = -\frac{1}{\sigma_i} A v_i = -u_i$.

By normalizing these vectors, we obtain the orthonormal eigenvectors of B.

$$q_i^+ = \frac{1}{\sqrt{2}} \begin{bmatrix} v_i \\ u_i \end{bmatrix}, \quad q_i^- = \frac{1}{\sqrt{2}} \begin{bmatrix} v_i \\ -u_i \end{bmatrix}$$

We can now assemble the full eigenvalue decomposition $B = Q\Lambda Q^T$.

The $2m \times 2m$ diagonal matrix of eigenvalues Λ contains the singular values and their negatives.

$$\Lambda = \begin{bmatrix} \Sigma & 0 \\ 0 & -\Sigma \end{bmatrix}$$

The corresponding orthogonal matrix Q is formed by concatenating the normalized eigenvectors.

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} V & V \\ U & -U \end{bmatrix}$$

The orthogonality of Q is verified by the following product.

$$Q^T Q = \frac{1}{2} \begin{bmatrix} V^T & U^T \\ V^T & -U^T \end{bmatrix} \begin{bmatrix} V & V \\ U & -U \end{bmatrix} = \frac{1}{2} \begin{bmatrix} V^T V + U^T U & V^T V - U^T U \\ V^T V - U^T U & V^T V + U^T U \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

Therefore, the eigenvalue decomposition of matrix B is given by the following expression.

$$B = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} V & V \\ U & -U \end{bmatrix} \right) \begin{bmatrix} \Sigma & 0 \\ 0 & -\Sigma \end{bmatrix} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} V^T & U^T \\ V^T & -U^T \end{bmatrix} \right)$$

Prob. 14

(1)

If A is a Hermitian matrix, it satisfies the property $A = A^*$, where A^* denotes the conjugate transpose. According to the spectral theorem, there exists a unitary matrix U and a diagonal matrix of eigenvalues Λ such that $A = U\Lambda U^*$. The induced 2-norm of a matrix is defined by the following expression.

$$\|A\|_2 = \sqrt{\lambda_{max}(A^* A)}$$

Substituting the identity for Hermitian matrices, we have $A^* A = A^2$. Using the diagonalization of A, we can express A^2 in terms of the eigenvalues.

$$A^2 = (U\Lambda U^*)(U\Lambda U^*) = U\Lambda^2 U^*$$

The eigenvalues of A^2 are exactly the squares of the eigenvalues of A, denoted by λ_i^2. Consequently, the maximum eigenvalue of $A^* A$ is given by the following.

$$\lambda_{max}(A^* A) = \max_i |\lambda_i|^2 = \left(\max_i |\lambda_i| \right)^2 = (\rho(A))^2$$

Taking the square root of both sides, we reach the desired equality.

$$\|A\|_2 = \rho(A)$$

(2)

Let λ be an eigenvalue of A such that its magnitude equals the spectral radius, $|\lambda| = \rho(A)$. There exists a corresponding non-zero eigenvector x that satisfies the eigenvalue equation.

$$Ax = \lambda x$$

By the definition of an induced matrix norm, the following inequality must hold for any vector x.

$$\|Ax\| \leq \|A\| \|x\|$$

Substituting the eigenvalue equation into the norm on the left-hand side, we get the following.

$$\|\lambda x\| \leq \|A\|\|x\|$$

Using the absolute homogeneity property of norms, we pull the scalar λ outside.

$$|\lambda|\|x\| \leq \|A\|\|x\|$$

Since x is a non-zero vector, we have $\|x\| > 0$, allowing us to divide both sides by $\|x\|$. This directly yields the inequality for the spectral radius.

$$\rho(A) \leq \|A\|$$

(3)

To prove $\lim_{k\to\infty} \|A^k\|^{1/k} = \rho(A)$, we establish both a lower and an upper bound. First, we apply the result from the previous section to the matrix A^k.

$$\rho(A^k) \leq \|A^k\|$$

Since the eigenvalues of A^k are λ_i^k, it follows that $\rho(A^k) = (\rho(A))^k$.

$$\rho(A)^k \leq \|A^k\| \implies \rho(A) \leq \|A^k\|^{1/k}$$

For the upper bound, let $\epsilon > 0$ be an arbitrary constant. According to the first provided fact, there exists a specific norm $\|\cdot\|_a$ such that $\|A\|_a \leq \rho(A) + \epsilon$. By the submultiplicative property of induced norms, we have the following bound for A^k.

$$\|A^k\|_a \leq (\|A\|_a)^k \leq (\rho(A) + \epsilon)^k$$

Using the second fact regarding norm equivalence, there exists a constant $c > 0$ such that our target norm $\|\cdot\|$ is bounded.

$$\|A^k\| \leq c\|A^k\|_a \leq c(\rho(A) + \epsilon)^k$$

Taking the k-th root of the entire expression, we obtain the following.

$$\|A^k\|^{1/k} \leq c^{1/k}(\rho(A) + \epsilon)$$

As k approaches infinity, the term $c^{1/k}$ converges to 1, leading to the limit bound.

$$\limsup_{k\to\infty} \|A^k\|^{1/k} \leq \rho(A) + \epsilon$$

Since this holds for any $\epsilon > 0$, we conclude that the limit equals the spectral radius.

$$\lim_{k\to\infty} \|A^k\|^{1/k} = \rho(A)$$

(4)

Suppose the spectral radius of A is strictly less than 1. We can choose a sufficiently small $\epsilon > 0$ such that $\rho(A) + \epsilon = q < 1$. Based on the facts used in Gelfand's formula, there exists a norm $\|\cdot\|_a$ such that the following is true.

$$\|A\|_a \leq \rho(A) + \epsilon = q$$

Considering the norm of the k-th power of A in this specific norm space, we have the following inequality.

$$\|A^k\|_a \leq (\|A\|_a)^k \leq q^k$$

Because $0 \leq q < 1$, the sequence q^k vanishes as k goes to infinity.

$$\lim_{k\to\infty} \|A^k\|_a = 0$$

By the principle of norm equivalence, all norms on a finite-dimensional space are equivalent, including the Frobenius norm.

$$\|A^k\|_F \leq c\|A^k\|_a$$

As the right-hand side goes to zero, the Frobenius norm of A^k must also converge to zero.

$$\lim_{k\to\infty} \|A^k\|_F = 0$$

Since the convergence of the Frobenius norm to zero implies that every individual entry of the matrix converges to zero, we conclude the final result.

$$\lim_{k\to\infty} A^k = 0$$